

Matrices & Vectors

ANSWER KEY



Addition and scalar multiplication

- 1 Let A be the matrix with rows $\langle 1, 3 \rangle$ and $\langle -2, 4 \rangle$, and B be the matrix with rows $\langle 2, -1 \rangle$ and $\langle 0, 5 \rangle$. Find $A + B$ and $2A$.

$A + B$ has rows $\langle 3, 2 \rangle$ and $\langle -2, 9 \rangle$. $2A$ has rows $\langle 2, 6 \rangle$ and $\langle -4, 8 \rangle$.

Matrix multiplication

- 2 Using A (rows $\langle 1, 3 \rangle$, $\langle -2, 4 \rangle$) and B (rows $\langle 2, -1 \rangle$, $\langle 0, 5 \rangle$) from above, find AB .

AB has rows $\langle 2, 14 \rangle$ and $\langle -4, 22 \rangle$: e.g. row 1, column 1 is $(1)(2) + (3)(0) = 2$.

- 3 Let A have rows $\langle 1, 3 \rangle$ and $\langle -2, 4 \rangle$, and let $\mathbf{v} = \langle 3, -1 \rangle$. Find $A\mathbf{v}$.

$A\mathbf{v} = \langle (1)(3) + (3)(-1), (-2)(3) + (4)(-1) \rangle = \langle 0, -10 \rangle$.

Determinant and inverse

- 4 Find $\det A$ for the matrix A with rows $\langle 1, 3 \rangle$ and $\langle -2, 4 \rangle$, and state whether A is invertible.

$\det A = (1)(4) - (3)(-2) = 4 + 6 = 10 \neq 0$, so A is invertible.

- 5 Find A^{-1} for the matrix A with rows $\langle 1, 3 \rangle$ and $\langle -2, 4 \rangle$.

$A^{-1} = \frac{1}{10} \times$ the matrix with rows $\langle 4, -3 \rangle$ and $\langle 2, 1 \rangle$, i.e., A^{-1} has rows $\langle 0.4, -0.3 \rangle$ and $\langle 0.2, 0.1 \rangle$.

Solving systems with the inverse matrix

- 6 Solve the system $x + 3y = 13$ and $-2x + 4y = 4$ using $\mathbf{x} = A^{-1}\mathbf{b}$ (using A^{-1} from above).

A^{-1} has rows $\langle 0.4, -0.3 \rangle$ and $\langle 0.2, 0.1 \rangle$. Then $x = 0.4(13) - 0.3(4) = 5.2 - 1.2 = 4$ and $y = 0.2(13) + 0.1(4) = 2.6 + 0.4 = 3$. Check: $4 + 3(3) = 13$ and $-2(4) + 4(3) = 4$.

Vector magnitude and dot product

- 7 For $\mathbf{u} = \langle 4, 3 \rangle$ and $\mathbf{w} = \langle -3, 4 \rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.

$\mathbf{u} \cdot \mathbf{w} = (4)(-3) + (3)(4) = 0$, and $|\mathbf{u}| = \sqrt{4^2 + 3^2} = 5$.

Angle between vectors

- 8 Find the angle between $\mathbf{p} = \langle 2, 2 \rangle$ and $\mathbf{q} = \langle 3, 0 \rangle$.

$$\mathbf{p} \cdot \mathbf{q} = 6, |\mathbf{p}| = 2\sqrt{2}, |\mathbf{q}| = 3. \cos \theta = \frac{6}{6\sqrt{2}} = \frac{1}{\sqrt{2}} \approx 0.7071 \Rightarrow \theta = 45^\circ.$$

- 9 Find the angle between $\langle 7, 1 \rangle$ and $\langle 2, -3 \rangle$, rounded to the nearest tenth of a degree.

$$\text{Dot product} = 14 - 3 = 11, \text{magnitudes } \sqrt{50} \text{ and } \sqrt{13}. \cos \theta = \frac{11}{\sqrt{650}} \approx 0.4314 \Rightarrow \theta \approx 64.4^\circ.$$

Multiple choice — easy

- 10 Let A be the matrix with rows $\langle -8, 4 \rangle$ and $\langle 7, 7 \rangle$, and B be the matrix with rows $\langle -2, 4 \rangle$ and $\langle -7, -3 \rangle$. Find $A + B$.

- a. $\langle -8, 8 \rangle, \langle 0, 4 \rangle$
- b. $\langle -10, 8 \rangle, \langle 0, 2 \rangle$
- c. $\langle -10, 8 \rangle, \langle 0, 4 \rangle$ ✓
- d. $\langle 0, 4 \rangle, \langle -10, 8 \rangle$

$A + B$ has rows $\langle -10, 8 \rangle$ and $\langle 0, 4 \rangle$.

- 11 Let A be the matrix with rows $\langle -4, 8 \rangle$ and $\langle -9, -2 \rangle$, and B be the matrix with rows $\langle -3, -8 \rangle$ and $\langle 5, -1 \rangle$. Find $A + B$.

- a. $\langle -7, 0 \rangle, \langle -4, -3 \rangle$ ✓
- b. $\langle -7, 0 \rangle, \langle -4, -5 \rangle$
- c. $\langle -5, 0 \rangle, \langle -4, -3 \rangle$
- d. $\langle -4, -3 \rangle, \langle -7, 0 \rangle$

$A + B$ has rows $\langle -7, 0 \rangle$ and $\langle -4, -3 \rangle$.

- 12** Let A be the matrix with rows $\langle -8, -2 \rangle$ and $\langle 8, 8 \rangle$, and B be the matrix with rows $\langle -4, -8 \rangle$ and $\langle -5, 5 \rangle$. Find $A + B$.

- a. $\langle 3, 13 \rangle, \langle -12, -10 \rangle$
- b. $\langle -10, -10 \rangle, \langle 3, 13 \rangle$
- c. $\langle -12, -10 \rangle, \langle 3, 13 \rangle$ ✓
- d. $\langle -12, -10 \rangle, \langle 3, 11 \rangle$

$A + B$ has rows $\langle -12, -10 \rangle$ and $\langle 3, 13 \rangle$.

- 13** Let A be the matrix with rows $\langle 7, 4 \rangle$ and $\langle 3, 5 \rangle$, and B be the matrix with rows $\langle 9, -5 \rangle$ and $\langle -4, 3 \rangle$. Find $A + B$.

- a. $\langle 16, -1 \rangle, \langle -1, 6 \rangle$
- b. $\langle -1, 8 \rangle, \langle 16, -1 \rangle$
- c. $\langle 18, -1 \rangle, \langle -1, 8 \rangle$
- d. $\langle 16, -1 \rangle, \langle -1, 8 \rangle$ ✓

$A + B$ has rows $\langle 16, -1 \rangle$ and $\langle -1, 8 \rangle$.

- 14** Let A be the matrix with rows $\langle 1, -3 \rangle$ and $\langle 7, 6 \rangle$, and B be the matrix with rows $\langle 6, 2 \rangle$ and $\langle -5, 1 \rangle$. Find $A + B$.

- a. $\langle 7, -1 \rangle, \langle 2, 5 \rangle$
- b. $\langle 2, 7 \rangle, \langle 7, -1 \rangle$
- c. $\langle 7, -1 \rangle, \langle 2, 7 \rangle$ ✓
- d. $\langle 9, -1 \rangle, \langle 2, 7 \rangle$

$A + B$ has rows $\langle 7, -1 \rangle$ and $\langle 2, 7 \rangle$.

- 15** Let A be the matrix with rows $\langle -2, -6 \rangle$ and $\langle 5, 7 \rangle$, and B be the matrix with rows $\langle -7, 4 \rangle$ and $\langle 5, 7 \rangle$. Find $A + B$.

a. $\langle 10, 14 \rangle, \langle -9, -2 \rangle$

b. $\langle -9, -2 \rangle, \langle 10, 12 \rangle$

c. $\langle -9, -2 \rangle, \langle 10, 14 \rangle$ ✓

d. $\langle -7, -2 \rangle, \langle 10, 14 \rangle$

$A + B$ has rows $\langle -9, -2 \rangle$ and $\langle 10, 14 \rangle$.

- 16** Let A be the matrix with rows $\langle 6, -2 \rangle$ and $\langle -8, -6 \rangle$, and B be the matrix with rows $\langle -5, -4 \rangle$ and $\langle 5, -6 \rangle$. Find $A + B$.

a. $\langle 1, -6 \rangle, \langle -3, -12 \rangle$ ✓

b. $\langle 3, -6 \rangle, \langle -3, -12 \rangle$

c. $\langle -3, -12 \rangle, \langle 1, -6 \rangle$

d. $\langle 1, -6 \rangle, \langle -3, -14 \rangle$

$A + B$ has rows $\langle 1, -6 \rangle$ and $\langle -3, -12 \rangle$.

- 17** Let A be the matrix with rows $\langle -7, -9 \rangle$ and $\langle 1, -5 \rangle$, and B be the matrix with rows $\langle 7, -7 \rangle$ and $\langle -3, 6 \rangle$. Find $A + B$.

a. $\langle 0, -16 \rangle, \langle -2, -1 \rangle$

b. $\langle 2, -16 \rangle, \langle -2, 1 \rangle$

c. $\langle -2, 1 \rangle, \langle 0, -16 \rangle$

d. $\langle 0, -16 \rangle, \langle -2, 1 \rangle$ ✓

$A + B$ has rows $\langle 0, -16 \rangle$ and $\langle -2, 1 \rangle$.

- 18 Let A be the matrix with rows $\langle -3, -6 \rangle$ and $\langle -7, 1 \rangle$, and B be the matrix with rows $\langle -5, -1 \rangle$ and $\langle -4, 3 \rangle$. Find $A + B$.

- a. $\langle -6, -7 \rangle, \langle -11, 4 \rangle$
- b. $\langle -8, -7 \rangle, \langle -11, 4 \rangle$ ✓
- c. $\langle -8, -7 \rangle, \langle -11, 2 \rangle$
- d. $\langle -11, 4 \rangle, \langle -8, -7 \rangle$

$A + B$ has rows $\langle -8, -7 \rangle$ and $\langle -11, 4 \rangle$.

- 19 Let A be the matrix with rows $\langle -4, 3 \rangle$ and $\langle -7, 1 \rangle$, and B be the matrix with rows $\langle -2, 9 \rangle$ and $\langle -5, 1 \rangle$. Find $A + B$.

- a. $\langle -4, 12 \rangle, \langle -12, 2 \rangle$
- b. $\langle -6, 12 \rangle, \langle -12, 0 \rangle$
- c. $\langle -12, 2 \rangle, \langle -6, 12 \rangle$
- d. $\langle -6, 12 \rangle, \langle -12, 2 \rangle$ ✓

$A + B$ has rows $\langle -6, 12 \rangle$ and $\langle -12, 2 \rangle$.

- 20 Let A be the matrix with rows $\langle -6, 4 \rangle$ and $\langle -7, -9 \rangle$, and B be the matrix with rows $\langle 2, 6 \rangle$ and $\langle 1, -6 \rangle$. Find $A + B$.

- a. $\langle -4, 10 \rangle, \langle -6, -15 \rangle$ ✓
- b. $\langle -2, 10 \rangle, \langle -6, -15 \rangle$
- c. $\langle -4, 10 \rangle, \langle -6, -17 \rangle$
- d. $\langle -6, -15 \rangle, \langle -4, 10 \rangle$

$A + B$ has rows $\langle -4, 10 \rangle$ and $\langle -6, -15 \rangle$.

- 21** Let A be the matrix with rows $\langle 8, -7 \rangle$ and $\langle 9, -7 \rangle$, and B be the matrix with rows $\langle 7, 9 \rangle$ and $\langle 1, 9 \rangle$. Find $A + B$.

- a. $\langle 17, 2 \rangle, \langle 10, 2 \rangle$
- b. $\langle 10, 2 \rangle, \langle 15, 2 \rangle$
- c. $\langle 15, 2 \rangle, \langle 10, 2 \rangle$ ✓
- d. $\langle 15, 2 \rangle, \langle 10, 0 \rangle$

$A + B$ has rows $\langle 15, 2 \rangle$ and $\langle 10, 2 \rangle$.

- 22** Let A be the matrix with rows $\langle -3, -1 \rangle$ and $\langle 5, 1 \rangle$, and B be the matrix with rows $\langle -4, 4 \rangle$ and $\langle 9, -7 \rangle$. Find $A + B$.

- a. $\langle -5, 3 \rangle, \langle 14, -6 \rangle$
- b. $\langle -7, 3 \rangle, \langle 14, -6 \rangle$ ✓
- c. $\langle -7, 3 \rangle, \langle 14, -8 \rangle$
- d. $\langle 14, -6 \rangle, \langle -7, 3 \rangle$

$A + B$ has rows $\langle -7, 3 \rangle$ and $\langle 14, -6 \rangle$.

- 23** Let A be the matrix with rows $\langle -6, 5 \rangle$ and $\langle -9, 1 \rangle$, and B be the matrix with rows $\langle 8, 3 \rangle$ and $\langle 7, 6 \rangle$. Find $A + B$.

- a. $\langle 4, 8 \rangle, \langle -2, 7 \rangle$
- b. $\langle 2, 8 \rangle, \langle -2, 5 \rangle$
- c. $\langle 2, 8 \rangle, \langle -2, 7 \rangle$ ✓
- d. $\langle -2, 7 \rangle, \langle 2, 8 \rangle$

$A + B$ has rows $\langle 2, 8 \rangle$ and $\langle -2, 7 \rangle$.

- 24 Let A be the matrix with rows $\langle -9, 3 \rangle$ and $\langle 9, 3 \rangle$, and B be the matrix with rows $\langle 1, 9 \rangle$ and $\langle 3, -3 \rangle$. Find $A + B$.

- a. $\langle -6, 12 \rangle, \langle 12, 0 \rangle$
- b. $\langle -8, 12 \rangle, \langle 12, 0 \rangle$ ✓
- c. $\langle -8, 12 \rangle, \langle 12, -2 \rangle$
- d. $\langle 12, 0 \rangle, \langle -8, 12 \rangle$

$A + B$ has rows $\langle -8, 12 \rangle$ and $\langle 12, 0 \rangle$.

Multiple choice — medium

- 25 Let A have rows $\langle -9, 8 \rangle$ and $\langle -9, 6 \rangle$, and let $\mathbf{v} = \langle 6, 3 \rangle$. Find $A\mathbf{v}$.

- a. $\langle -36, -30 \rangle$
- b. $\langle -28, -36 \rangle$
- c. $\langle -30, -36 \rangle$ ✓
- d. $\langle -30, -38 \rangle$

$A\mathbf{v} = \langle (-9)(6) + (8)(3), (-9)(6) + (6)(3) \rangle = \langle -30, -36 \rangle$.

- 26 Let A have rows $\langle -1, -3 \rangle$ and $\langle 9, -6 \rangle$, and let $\mathbf{v} = \langle -5, 1 \rangle$. Find $A\mathbf{v}$.

- a. $\langle 2, -51 \rangle$ ✓
- b. $\langle -51, 2 \rangle$
- c. $\langle 4, -51 \rangle$
- d. $\langle 2, -53 \rangle$

$A\mathbf{v} = \langle (-1)(-5) + (-3)(1), (9)(-5) + (-6)(1) \rangle = \langle 2, -51 \rangle$.

27 Let A have rows $\langle 7, 4 \rangle$ and $\langle 1, 4 \rangle$, and let $\mathbf{v} = \langle -5, 5 \rangle$. Find $A\mathbf{v}$.

a. $\langle -15, 13 \rangle$

b. $\langle 15, -15 \rangle$

c. $\langle -15, 15 \rangle$ ✓

d. $\langle -13, 15 \rangle$

$$A\mathbf{v} = \langle (7)(-5) + (4)(5), (1)(-5) + (4)(5) \rangle = \langle -15, 15 \rangle.$$

28 Let A have rows $\langle 9, -1 \rangle$ and $\langle 4, -1 \rangle$, and let $\mathbf{v} = \langle 3, -5 \rangle$. Find $A\mathbf{v}$.

a. $\langle 17, 32 \rangle$

b. $\langle 32, 15 \rangle$

c. $\langle 34, 17 \rangle$

d. $\langle 32, 17 \rangle$ ✓

$$A\mathbf{v} = \langle (9)(3) + (-1)(-5), (4)(3) + (-1)(-5) \rangle = \langle 32, 17 \rangle.$$

29 Let A have rows $\langle -1, 8 \rangle$ and $\langle 2, 7 \rangle$, and let $\mathbf{v} = \langle 4, -8 \rangle$. Find $A\mathbf{v}$.

a. $\langle -48, -68 \rangle$

b. $\langle -66, -48 \rangle$

c. $\langle -68, -50 \rangle$

d. $\langle -68, -48 \rangle$ ✓

$$A\mathbf{v} = \langle (-1)(4) + (8)(-8), (2)(4) + (7)(-8) \rangle = \langle -68, -48 \rangle.$$

30 Let A have rows $\langle -4, 7 \rangle$ and $\langle 6, 4 \rangle$, and let $\mathbf{v} = \langle -8, 5 \rangle$. Find $A\mathbf{v}$.

a. $\langle 69, -28 \rangle$

b. $\langle 67, -28 \rangle$ ✓

c. $\langle -28, 67 \rangle$

d. $\langle 67, -30 \rangle$

$$A\mathbf{v} = \langle (-4)(-8) + (7)(5), (6)(-8) + (4)(5) \rangle = \langle 67, -28 \rangle.$$

31 Let A have rows $\langle -1, 5 \rangle$ and $\langle -2, 2 \rangle$, and let $\mathbf{v} = \langle 2, -6 \rangle$. Find $A\mathbf{v}$.

a. $\langle -32, -18 \rangle$

b. $\langle -16, -32 \rangle$

c. $\langle -32, -16 \rangle$ ✓

d. $\langle -30, -16 \rangle$

$$A\mathbf{v} = \langle (-1)(2) + (5)(-6), (-2)(2) + (2)(-6) \rangle = \langle -32, -16 \rangle.$$

32 Let A have rows $\langle -2, 2 \rangle$ and $\langle 8, -1 \rangle$, and let $\mathbf{v} = \langle -9, -1 \rangle$. Find $A\mathbf{v}$.

a. $\langle 18, -71 \rangle$

b. $\langle -71, 16 \rangle$

c. $\langle 16, -73 \rangle$

d. $\langle 16, -71 \rangle$ ✓

$$A\mathbf{v} = \langle (-2)(-9) + (2)(-1), (8)(-9) + (-1)(-1) \rangle = \langle 16, -71 \rangle.$$

33 Let A have rows $\langle 4, -6 \rangle$ and $\langle -7, -8 \rangle$, and let $\mathbf{v} = \langle -4, -4 \rangle$. Find $A\mathbf{v}$.

a. $\langle 8, 60 \rangle$ ✓

b. $\langle 60, 8 \rangle$

c. $\langle 10, 60 \rangle$

d. $\langle 8, 58 \rangle$

$$A\mathbf{v} = \langle (4)(-4) + (-6)(-4), (-7)(-4) + (-8)(-4) \rangle = \langle 8, 60 \rangle.$$

34 Let A have rows $\langle -5, -6 \rangle$ and $\langle 9, 6 \rangle$, and let $\mathbf{v} = \langle -7, -4 \rangle$. Find $A\mathbf{v}$.

a. $\langle 59, -87 \rangle$ ✓

b. $\langle -87, 59 \rangle$

c. $\langle 59, -89 \rangle$

d. $\langle 61, -87 \rangle$

$$A\mathbf{v} = \langle (-5)(-7) + (-6)(-4), (9)(-7) + (6)(-4) \rangle = \langle 59, -87 \rangle.$$

35 Let A have rows $\langle 4, -6 \rangle$ and $\langle 8, -7 \rangle$, and let $\mathbf{v} = \langle 6, -9 \rangle$. Find $A\mathbf{v}$.

a. $\langle 111, 78 \rangle$

b. $\langle 78, 111 \rangle$ ✓

c. $\langle 80, 111 \rangle$

d. $\langle 78, 109 \rangle$

$$A\mathbf{v} = \langle (4)(6) + (-6)(-9), (8)(6) + (-7)(-9) \rangle = \langle 78, 111 \rangle.$$

36 Let A have rows $\langle -5, -3 \rangle$ and $\langle 1, -2 \rangle$, and let $\mathbf{v} = \langle 9, 4 \rangle$. Find $A\mathbf{v}$.

a. $\langle -57, 1 \rangle$ ✓

b. $\langle -55, 1 \rangle$

c. $\langle 1, -57 \rangle$

d. $\langle -57, -1 \rangle$

$$A\mathbf{v} = \langle (-5)(9) + (-3)(4), (1)(9) + (-2)(4) \rangle = \langle -57, 1 \rangle.$$

37 Let A have rows $\langle -3, 4 \rangle$ and $\langle -1, -6 \rangle$, and let $\mathbf{v} = \langle 1, -4 \rangle$. Find $A\mathbf{v}$.

a. $\langle -19, 21 \rangle$

b. $\langle -17, 23 \rangle$

c. $\langle -19, 23 \rangle$ ✓

d. $\langle 23, -19 \rangle$

$$A\mathbf{v} = \langle (-3)(1) + (4)(-4), (-1)(1) + (-6)(-4) \rangle = \langle -19, 23 \rangle.$$

38 Let A have rows $\langle 4, 9 \rangle$ and $\langle -8, 6 \rangle$, and let $\mathbf{v} = \langle -7, 3 \rangle$. Find $A\mathbf{v}$.

- a. $\langle 1, 74 \rangle$
- b. $\langle 74, -1 \rangle$
- c. $\langle -1, 72 \rangle$
- d. $\langle -1, 74 \rangle$ ✓

$$A\mathbf{v} = \langle (4)(-7) + (9)(3), (-8)(-7) + (6)(3) \rangle = \langle -1, 74 \rangle.$$

39 Let A have rows $\langle 2, -6 \rangle$ and $\langle 3, 3 \rangle$, and let $\mathbf{v} = \langle 3, 8 \rangle$. Find $A\mathbf{v}$.

- a. $\langle -42, 31 \rangle$
- b. $\langle -42, 33 \rangle$ ✓
- c. $\langle 33, -42 \rangle$
- d. $\langle -40, 33 \rangle$

$$A\mathbf{v} = \langle (2)(3) + (-6)(8), (3)(3) + (3)(8) \rangle = \langle -42, 33 \rangle.$$

Multiple choice — hard

40 Find $\det A$ for the matrix A with rows $\langle -6, 5 \rangle$ and $\langle 7, -8 \rangle$.

- a. 13 ✓
- b. 15
- c. 11
- d. -13

$$\det A = (-6)(-8) - (5)(7) = 48 - 35 = 13, \text{ so } A \text{ is invertible since } \det A \neq 0.$$

41 Find $\det A$ for the matrix A with rows $\langle -5, -3 \rangle$ and $\langle 6, 4 \rangle$.

a. -2 ✓

b. 2

c. -4

d. 0

$\det A = (-5)(4) - (-3)(6) = -20 - (-18) = -2$, so A is invertible since $\det A \neq 0$.

42 Find $\det A$ for the matrix A with rows $\langle 7, 4 \rangle$ and $\langle 3, 5 \rangle$.

a. -23

b. 21

c. 25

d. 23 ✓

$\det A = (7)(5) - (4)(3) = 35 - 12 = 23$, so A is invertible since $\det A \neq 0$.

43 Find $\det A$ for the matrix A with rows $\langle 1, 8 \rangle$ and $\langle -1, 6 \rangle$.

a. -14

b. 12

c. 14 ✓

d. 16

$\det A = (1)(6) - (8)(-1) = 6 - (-8) = 14$, so A is invertible since $\det A \neq 0$.

44 Find $\det A$ for the matrix A with rows $\langle -3, -3 \rangle$ and $\langle -3, 5 \rangle$.

a. -26

b. 24

c. -22

d. -24 ✓

$\det A = (-3)(5) - (-3)(-3) = -15 - 9 = -24$, so A is invertible since $\det A \neq 0$.

45 For $\mathbf{u} = \langle -12, -35 \rangle$ and $\mathbf{w} = \langle 8, 1 \rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.

a. $-129, 37$

b. $-131, 37$ ✓

c. $-131, 39$

d. $37, -131$

$$\mathbf{u} \cdot \mathbf{w} = (-12)(8) + (-35)(1) = -131, \text{ and } |\mathbf{u}| = \sqrt{(-12)^2 + (-35)^2} = 37.$$

46 For $\mathbf{u} = \langle -20, -21 \rangle$ and $\mathbf{w} = \langle -8, 7 \rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.

a. $15, 29$

b. $13, 31$

c. $29, 13$

d. $13, 29$ ✓

$$\mathbf{u} \cdot \mathbf{w} = (-20)(-8) + (-21)(7) = 13, \text{ and } |\mathbf{u}| = \sqrt{(-20)^2 + (-21)^2} = 29.$$

47 For $\mathbf{u} = \langle -20, 21 \rangle$ and $\mathbf{w} = \langle -7, -5 \rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.

a. $29, 35$

b. $37, 29$

c. $35, 29$ ✓

d. $35, 31$

$$\mathbf{u} \cdot \mathbf{w} = (-20)(-7) + (21)(-5) = 35, \text{ and } |\mathbf{u}| = \sqrt{(-20)^2 + 21^2} = 29.$$

48 For $\mathbf{u} = \langle -12, 35 \rangle$ and $\mathbf{w} = \langle -7, 3 \rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.

- a. 191, 37
- b. 189, 37 ✓
- c. 37, 189
- d. 189, 39

$$\mathbf{u} \cdot \mathbf{w} = (-12)(-7) + (35)(3) = 189, \text{ and } |\mathbf{u}| = \sqrt{(-12)^2 + 35^2} = 37.$$

49 For $\mathbf{u} = \langle 9, -40 \rangle$ and $\mathbf{w} = \langle -3, -7 \rangle$, find $\mathbf{u} \cdot \mathbf{w}$ and $|\mathbf{u}|$.

- a. 41, 253
- b. 253, 41 ✓
- c. 253, 43
- d. 255, 41

$$\mathbf{u} \cdot \mathbf{w} = (9)(-3) + (-40)(-7) = 253, \text{ and } |\mathbf{u}| = \sqrt{9^2 + (-40)^2} = 41.$$

50 Find the angle between $\langle 4, -8 \rangle$ and $\langle -1, 2 \rangle$, rounded to the nearest tenth of a degree.

- a. 150.0°
- b. 165.0°
- c. 175.0°
- d. 180.0° ✓

$$\text{Dot product} = -20, \text{ magnitudes } \sqrt{80} \text{ and } \sqrt{5}. \cos \theta = \frac{-20}{\sqrt{400}} \approx -1.0000 \Rightarrow \theta \approx 180.0^\circ.$$

51 Find the angle between $\langle -5, 5 \rangle$ and $\langle 5, 2 \rangle$, rounded to the nearest tenth of a degree.

- a. 123.2°
- b. 113.2° ✓
- c. 108.2°
- d. 118.2°

$$\text{Dot product} = -15, \text{ magnitudes } \sqrt{50} \text{ and } \sqrt{29}. \cos \theta = \frac{-15}{\sqrt{1450}} \approx -0.3939 \Rightarrow \theta \approx 113.2^\circ.$$

52 Find the angle between $\langle 4, 8 \rangle$ and $\langle 3, 9 \rangle$, rounded to the nearest tenth of a degree.

a. 13.1°

b. 8.1° ✓

c. 3.1°

d. 18.1°

Dot product = 84, magnitudes $\sqrt{80}$ and $\sqrt{90}$. $\cos \theta = \frac{84}{\sqrt{7200}} \approx 0.9899 \Rightarrow \theta \approx 8.1^\circ$.

53 Find the angle between $\langle 4, 6 \rangle$ and $\langle -6, 1 \rangle$, rounded to the nearest tenth of a degree.

a. 124.2°

b. 119.2°

c. 114.2° ✓

d. 109.2°

Dot product = -18 , magnitudes $\sqrt{52}$ and $\sqrt{37}$. $\cos \theta = \frac{-18}{\sqrt{1924}} \approx -0.4104 \Rightarrow \theta \approx 114.2^\circ$.

54 Find the angle between $\langle 1, -2 \rangle$ and $\langle -2, -7 \rangle$, rounded to the nearest tenth of a degree.

a. 42.5° ✓

b. 47.5°

c. 52.5°

d. 37.5°

Dot product = 12, magnitudes $\sqrt{5}$ and $\sqrt{53}$. $\cos \theta = \frac{12}{\sqrt{265}} \approx 0.7372 \Rightarrow \theta \approx 42.5^\circ$.